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    " 'Doritos Corn Chips  Cheese Supreme 170g'\n",
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    " 'WW Original Corn    Chips 200g' 'Thins Potato Chips  Hot & Spicy 175g'\n",
    " 'Cobs Popd Sour Crm  &Chives Chips 110g'\n",
    " 'Smiths Crnkle Chip  Orgnl Big Bag 380g'\n",
    " 'Doritos Corn Chips  Nacho Cheese 170g'\n",
    " 'Kettle Sensations   BBQ&Maple 150g' 'WW D/Style Chip     Sea Salt 200g'\n",
    " 'Pringles Chicken    Salt Crips 134g' 'WW Original Stacked Chips 160g'\n",
    " 'Smiths Chip Thinly  CutSalt/Vinegr175g' 'Cheezels Cheese 330g'\n",
    " 'Tostitos Lightly    Salted 175g' 'Thins Chips Salt &  Vinegar 175g'\n",
    " 'Smiths Crinkle Cut  Chips Barbecue 170g' 'Cheetos Puffs 165g'\n",
    " 'RRD Sweet Chilli &  Sour Cream 165g' 'WW Crinkle Cut      Original 175g'\n",
    " 'Tostitos Splash Of  Lime 175g' 'Woolworths Medium   Salsa 300g'\n",
    " 'Kettle Tortilla ChpsBtroot&Ricotta 150g' 'CCs Tasty Cheese    175g'\n",
    " 'Woolworths Cheese   Rings 190g' 'Tostitos Smoked     Chipotle 175g'\n",
    " 'Pringles Barbeque   134g' 'WW Supreme Cheese   Corn Chips 200g'\n",
    " 'Pringles Mystery    Flavour 134g'\n",
    " 'Tyrrells Crisps     Ched & Chives 165g'\n",
    " 'Snbts Whlgrn Crisps Cheddr&Mstrd 90g' 'Cheetos Chs & Bacon Balls 190g'\n",
    " 'Pringles Slt Vingar 134g' 'Infuzions SourCream&Herbs Veg Strws 110g'\n",
    " 'Kettle Tortilla ChpsFeta&Garlic 150g'\n",
    " 'Infuzions Mango     Chutny Papadums 70g'\n",
    " 'RRD Steak &         Chimuchurri 150g' 'RRD Honey Soy       Chicken 165g'\n",
    " 'Sunbites Whlegrn    Crisps Frch/Onin 90g' 'RRD Salt & Vinegar  165g'\n",
    " 'Doritos Cheese      Supreme 330g' 'Smiths Crinkle Cut  Snag&Sauce 150g'\n",
    " 'WW Sour Cream &OnionStacked Chips 160g' 'RRD Lime & Pepper   165g'\n",
    " 'Natural ChipCo Sea  Salt & Vinegr 175g'\n",
    " 'Red Rock Deli Chikn&Garlic Aioli 150g'\n",
    " 'RRD SR Slow Rst     Pork Belly 150g' 'RRD Pc Sea Salt     165g'\n",
    " 'Smith Crinkle Cut   Bolognese 150g' 'Doritos Salsa Mild  300g']\n",
    "PROD_NAME: 114\n"
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175g'\n",
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      " 'smiths chip thinly  s/cream&onion 175g'\n",
      " 'kettle tortilla chpshny&jlpno chili 150g'\n",
      " 'old el paso salsa   dip tomato mild 300g'\n",
      " 'smiths crinkle chips salt & vinegar 330g'\n",
      " 'grain waves         sweet chilli 210g'\n",
      " 'doritos corn chip mexican jalapeno 150g'\n",
      " 'grain waves sour    cream&chives 210g'\n",
      " 'kettle sensations   siracha lime 150g' 'twisties cheese     270g'\n",
      " 'ww crinkle cut      chicken 175g' 'thins chips light&  tangy 175g'\n",
      " 'ccs original 175g' 'burger rings 220g'\n",
      " 'ncc sour cream &    garden chives 175g'\n",
      " 'doritos corn chip southern chicken 150g' 'cheezels cheese box 125g'\n",
      " 'smiths crinkle      original 330g'\n",
      " 'infzns crn crnchers tangy gcamole 110g'\n",
      " 'kettle sea salt     and vinegar 175g'\n",
      " 'smiths chip thinly  cut original 175g' 'kettle original 175g'\n",
      " 'red rock deli thai  chilli&lime 150g' 'pringles sthrn friedchicken 134g'\n",
      " 'pringles sweet&spcy bbq 134g' 'red rock deli sr    salsa & mzzrlla 150g'\n",
      " 'thins chips         originl saltd 175g'\n",
      " 'red rock deli sp    salt & truffle 150g'\n",
      " 'smiths thinly       swt chli&s/cream175g' 'kettle chilli 175g'\n",
      " 'doritos mexicana    170g' 'smiths crinkle cut  french oniondip 150g'\n",
      " 'natural chipco      hony soy chckn175g'\n",
      " 'dorito corn chp     supreme 380g' 'twisties chicken270g'\n",
      " 'smiths thinly cut   roast chicken 175g'\n",
      " 'smiths crinkle cut  tomato salsa 150g'\n",
      " 'kettle mozzarella   basil & pesto 175g'\n",
      " 'infuzions thai sweetchili potatomix 110g'\n",
      " 'kettle sensations   camembert & fig 150g'\n",
      " 'smith crinkle cut   mac n cheese 150g'\n",
      " 'kettle honey soy    chicken 175g' 'thins chips seasonedchicken 175g'\n",
      " 'smiths crinkle cut  salt & vinegar 170g'\n",
      " 'infuzions bbq rib   prawn crackers 110g'\n",
      " 'grnwves plus btroot & chilli jam 180g'\n",
      " 'tyrrells crisps     lightly salted 165g'\n",
      " 'kettle sweet chilli and sour cream 175g'\n",
```

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    " 'doritos salsa       medium 300g' 'kettle 135g swt pot sea salt'\n",
    " 'pringles sourcream  onion 134g' 'doritos corn chips  original 170g'\n",
    " 'twisties cheese     burger 250g'\n",
    " 'old el paso salsa   dip chnky tom ht300g'\n",
    " 'cobs popd swt/chlli &sr/cream chips 110g'\n",
    " 'woolworths mild     salsa 300g'\n",
    " 'natural chip co     tmato hrb&spce 175g'\n",
    " 'smiths crinkle cut  chips original 170g'\n",
    " 'cobs popd sea salt  chips 110g'\n",
    " 'smiths crinkle cut  chips chs&onion170g'\n",
    " 'french fries potato chips 175g'\n",
    " 'old el paso salsa   dip tomato med 300g'\n",
    " 'doritos corn chips  cheese supreme 170g'\n",
    " 'pringles original   crisps 134g' 'rrd chilli&         coconut 150g'\n",
    " 'ww original corn    chips 200g' 'thins potato chips  hot & spicy 175g'\n",
    " 'cobs popd sour crm  &chives chips 110g'\n",
    " 'smiths crnkle chip  orgnl big bag 380g'\n",
    " 'doritos corn chips  nacho cheese 170g'\n",
    " 'kettle sensations   bbq&maple 150g' 'ww d/style chip     sea salt 200g'\n",
    " 'pringles chicken    salt crips 134g' 'ww original stacked chips 160g'\n",
    " 'smiths chip thinly  cutsalt/vinegr175g' 'cheezels cheese 330g'\n",
    " 'tostitos lightly    salted 175g' 'thins chips salt &  vinegar 175g'\n",
    " 'smiths crinkle cut  chips barbecue 170g' 'cheetos puffs 165g'\n",
    " 'rrd sweet chilli &  sour cream 165g' 'ww crinkle cut      original 175g'\n",
    " 'tostitos splash of  lime 175g' 'woolworths medium   salsa 300g'\n",
    " 'kettle tortilla chpsbtroot&ricotta 150g' 'ccs tasty cheese    175g'\n",
    " 'woolworths cheese   rings 190g' 'tostitos smoked     chipotle 175g'\n",
    " 'pringles barbeque   134g' 'ww supreme cheese   corn chips 200g'\n",
    " 'pringles mystery    flavour 134g'\n",
    " 'tyrrells crisps     ched & chives 165g'\n",
    " 'snbts whlgrn crisps cheddr&mstrd 90g' 'cheetos chs & bacon balls 190g'\n",
    " 'pringles slt vingar 134g' 'infuzions sourcream&herbs veg strws 110g'\n",
    " 'kettle tortilla chpsfeta&garlic 150g'\n",
    " 'infuzions mango     chutny papadums 70g'\n",
    " 'rrd steak &         chimuchurri 150g' 'rrd honey soy       chicken 165g'\n",
    " 'sunbites whlegrn    crisps frch/onin 90g' 'rrd salt & vinegar  165g'\n",
    " 'doritos cheese      supreme 330g' 'smiths crinkle cut  snag&sauce 150g'\n",
    " 'ww sour cream &onionstacked chips 160g' 'rrd lime & pepper   165g'\n",
    " 'natural chipco sea  salt & vinegr 175g'\n",
    " 'red rock deli chikn&garlic aioli 150g'\n",
    " 'rrd sr slow rst     pork belly 150g' 'rrd pc sea salt     165g'\n",
    " 'smith crinkle cut   bolognese 150g' 'doritos salsa mild  300g']\n",
    "PROD_NAME: 114\n"
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    "4       kettle tortilla chpshny&jlpno chili 150g\n",
    "5       old el paso salsa   dip tomato mild 300g\n",
    "7          grain waves         sweet chilli 210g\n",
    "9          grain waves sour    cream&chives 210g\n",
    "...                                           ...\n",
    "264829  infuzions sourcream&herbs veg strws 110g\n",
    "264831   kettle sweet chilli and sour cream 175g\n",
    "264832             tostitos splash of  lime 175g\n",
    "264833                  doritos mexicana    170g\n",
    "264835             tostitos splash of  lime 175g\n",
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  "    pattern,\n",
  "    case = False,\n",
  "    na = False,\n",
  "    regex = True\n",
  ")\n",
  "\n",
  "df_no_chips = df_transactions[~mask_has_chip]\n",
  "\n",
  "print(df_no_chips[[\"PROD_NAME\"]])\n",
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" 'old el paso salsa dip tomato mild 300g'\n",
" 'grain waves sweet chilli 210g'\n",
" 'grain waves sour cream&chives 210g'\n",
" 'kettle sensations siracha lime 150g' 'twisties cheese 270g'\n",
" 'ww crinkle cut chicken 175g' 'ccs original 175g'\n",
" 'burger rings 220g' 'ncc sour cream & garden chives 175g'\n",
" 'cheezels cheese box 125g' 'smiths crinkle original 330g'\n",
" 'infzns crn crnchers tangy gcamole 110g'\n",
" 'kettle sea salt and vinegar 175g' 'kettle original 175g'\n",
" 'red rock deli thai chilli&lime 150g' 'pringles sthrn friedchicken 134g'\n",
" 'pringles sweet&spcy bbq 134g' 'red rock deli sr salsa & mzzrlla 150g'\n",
" 'red rock deli sp salt & truffle 150g'\n",
" 'smiths thinly swt chli&s/cream175g' 'kettle chilli 175g'\n",
" 'doritos mexicana 170g' 'smiths crinkle cut french oniondip 150g'\n",
" 'natural chipco hony soy chckn175g'\n",
" 'dorito corn chp supreme 380g' 'twisties chicken270g'\n",
" 'smiths thinly cut roast chicken 175g'\n",
" 'smiths crinkle cut tomato salsa 150g'\n",
" 'kettle mozzarella basil & pesto 175g'\n",
" 'infuzions thai sweetchili potatomix 110g'\n",
" 'kettle sensations camembert & fig 150g'\n",
" 'smith crinkle cut mac n cheese 150g'\n",
" 'kettle honey soy chicken 175g'\n",
" 'smiths crinkle cut salt & vinegar 170g'\n",
" 'infuzions bbq rib prawn crackers 110g'\n",
" 'grnwves plus btroot & chilli jam 180g'\n",
" 'tyrrells crisps lightly salted 165g'\n",
" 'kettle sweet chilli and sour cream 175g'\n",
" 'doritos salsa medium 300g' 'kettle 135g swt pot sea salt'\n",
" 'pringles sourcream onion 134g' 'twisties cheese burger 250g'\n",
" 'old el paso salsa dip chnky tom ht300g'\n",
" 'woolworths mild salsa 300g'\n",
" 'old el paso salsa dip tomato med 300g'\n",
" 'pringles original crisps 134g' 'rrd chilli& coconut 150g'\n",
" 'kettle sensations bbq&maple 150g' 'cheezels cheese 330g'\n",
" 'tostitos lightly salted 175g' 'cheetos puffs 165g'\n",
" 'rrd sweet chilli & sour cream 165g' 'ww crinkle cut original 175g'\n",
" 'tostitos splash of lime 175g' 'woolworths medium salsa 300g'\n",
" 'kettle tortilla chpsbtroot&ricotta 150g' 'ccs tasty cheese 175g'\n",
" 'woolworths cheese rings 190g' 'tostitos smoked chipotle 175g'\n",
" 'pringles barbeque 134g' 'pringles mystery flavour 134g'\n",
" 'tyrrells crisps ched & chives 165g'\n",
" 'snbts whlgrn crisps cheddr&mstrd 90g' 'cheetos chs & bacon balls 190g'\n",
" 'pringles slt vingar 134g' 'infuzions sourcream&herbs veg strws 110g'\n",
" 'kettle tortilla chpsfeta&garlic 150g'\n",
" 'infuzions mango chutny papadums 70g'\n",
" 'rrd steak & chimuchurri 150g' 'rrd honey soy chicken 165g'\n",
" 'sunbites whlegrn crisps frch/onin 90g' 'rrd salt & vinegar 165g'\n",
" 'doritos cheese supreme 330g' 'smiths crinkle cut snag&sauce 150g'\n",
" 'rrd lime & pepper 165g' 'natural chipco sea salt & vinegr 175g'\n",

```
      " 'red rock deli chikn&garlic aioli 150g'\n",
      " 'rrd sr slow rst     pork belly 150g' 'rrd pc sea salt     165g'\n",
      " 'smith crinkle cut   bolognese 150g' 'doritos salsa mild  300g']\n",
      "PROD_NAME: 83\n"
     ]
    }
   ],
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    "print(f'PROD_NAME: {df_no_chips[\"PROD_NAME\"].nunique()}')"
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      "7      grain waves         sweet chilli 210g\n",
      "9      grain waves sour    cream&chives 210g\n",
      "13          ww crinkle cut      chicken 175g\n",
      "16                         burger rings 220g\n",
      "17    ncc sour cream &    garden chives 175g\n",
      "21    ncc sour cream &    garden chives 175g\n",
      "22    infzns crn crnchers tangy gcamole 110g\n",
      "25  old el paso salsa   dip tomato mild 300g\n",
      "28      red rock deli thai  chilli&lime 150g\n",
      "29    infzns crn crnchers tangy gcamole 110g\n",
      "32  red rock deli sr    salsa & mzzrlla 150g\n",
      "34   red rock deli sp    salt & truffle 150g\n",
      "41          dorito corn chp     supreme 380g\n",
      "46    infzns crn crnchers tangy gcamole 110g\n",
      "47  infuzions thai sweetchili potatomix 110g\n",
      "50     smith crinkle cut   mac n cheese 150g\n",
      "55   infuzions bbq rib   prawn crackers 110g\n",
      "56     grnwves plus btroot & chilli jam 180g\n",
      "64  infuzions thai sweetchili potatomix 110g\n",
      "Total non-chip rows: 57954\n"
     ]
    }
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    "    \"smiths\", \"kettle\", \"doritos\", \"twisties\", \"pringles\",\n",
      "       \"tyrrells\",  \"cheezels\",  \"tostitos\",  \"snbts\",  \"rrd\",  \"natural
chipco\", \"ccs\"\n",
    "]\n",
```

```
    "\n",
    "pattern = r\"\\b(chip|chips|crip|crips|potato|potatoes)\\b\"\n",
    "\n",
    "df_transactions[\"is_chip\"] = (\n",
      "      df_transactions[\"PROD_NAME\"].str.contains(pattern,  case=False,  na=False,
regex=True) |\n",
     "     df_transactions[\"PROD_NAME\"].str.lower().str.contains(\"|\".join([b.lower()
for b in chip_brands]))\n",
    ")\n",
    "\n",
    "df_no_chips = df_transactions[~df_transactions[\"is_chip\"]]\n",
    "\n",
    "print(df_no_chips[[\"PROD_NAME\"]].head(20))\n",
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      "7        grain waves         sweet chilli 210g\n",
      "9        grain waves sour    cream&chives 210g\n",
      "16                           burger rings 220g\n",
      "17      ncc sour cream &    garden chives 175g\n",
      "21      ncc sour cream &    garden chives 175g\n",
      "22      infzns crn crnchers tangy gcamole 110g\n",
      "25    old el paso salsa   dip tomato mild 300g\n",
      "28        red rock deli thai  chilli&lime 150g\n",
      "29      infzns crn crnchers tangy gcamole 110g\n",
      "31                pringles sweet&spcy bbq 134g\n",
      "32    red rock deli sr    salsa & mzzrlla 150g\n",
      "34     red rock deli sp    salt & truffle 150g\n",
      "39    smiths crinkle cut  french oniondip 150g\n",
      "41            dorito corn chp     supreme 380g\n",
      "44       smiths crinkle cut  tomato salsa 150g\n",
      "46      infzns crn crnchers tangy gcamole 110g\n",
      "47    infuzions thai sweetchili potatomix 110g\n",
      "55     infuzions bbq rib   prawn crackers 110g\n",
      "56       grnwves plus btroot & chilli jam 180g\n",
      "Name: PROD_NAME, dtype: object\n",
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   "source": [
    "# Likely chip keywords\n",
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    "chip_keywords = [\n",
    "    \"chip\",\"chips\",\"crisp\",\"crisps\",\"potato\",\"potatoes\",\n",
    "    \"crinkle cut\",\"tortilla\",\"sensations\",\"original\"\n",
    "]\n",
    "\n",
    "# Known chip brands\n",
    "chip_brands = [\n",
    "    \"smiths\", \"kettle\", \"doritos\", \"twisties\", \"pringles\",\n",
       "       \"tyrrells\",  \"cheezels\",  \"tostitos\",  \"snbts\",  \"rrd\",  \"natural
chipco\", \"ccs\",\n",
    "]\n",
    "\n",
    "# Obvious non-chip categories\n",
    "non_chip_keywords = [\n",
                                "                                \"salsa\",\"dip\",\"prawn
crackers\",\"nuts\",\"papadums\",\"sweet\",\"chocolate\",\"cheese sticks\"\n",
    "]\n",
    "\n",
    "# Lowercase product names for consistent matching\n",
    "prod_lower = df_transactions[\"PROD_NAME\"].str.lower()\n",
    "\n",
    "# Step 1: Identify likely chips\n",
    "is_chip = prod_lower.str.contains(\"|\".join(chip_keywords), na=False) | \\\n",
     "             prod_lower.str.contains(\"|\".join([b.lower() for b in chip_brands]),
na=False)\n",
    "\n",
    "# Step 2: Identify obvious non-chips\n",
    "is_non_chip = prod_lower.str.contains(\"|\".join(non_chip_keywords), na=False)\n",
    "\n",
    "# Step 3: Final flag\n",
    "df_transactions[\"is_chip_final\"] = is_chip & (~is_non_chip)\n",
    "\n",
    "# Rows that are definitely not chips\n",
    "df_no_chips = df_transactions[~df_transactions[\"is_chip_final\"]]\n",
    "\n",
    "print(df_no_chips[\"PROD_NAME\"].head(20))\n",
    "print(f\"Total non-chip rows: {len(df_no_chips)}\")"
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      " 'grain waves         sweet chilli 210g'\n",
      " 'grain waves sour    cream&chives 210g' 'burger rings 220g'\n",
      " 'ncc sour cream &    garden chives 175g'\n",
```

```
      " 'infzns crn crnchers tangy gcamole 110g'\n",
      " 'red rock deli thai  chilli&lime 150g' 'pringles sweet&spcy bbq 134g'\n",
      " 'red rock deli sr    salsa & mzzrlla 150g'\n",
      " 'red rock deli sp    salt & truffle 150g'\n",
      " 'smiths crinkle cut  french oniondip 150g'\n",
      " 'dorito corn chp     supreme 380g'\n",
      " 'smiths crinkle cut  tomato salsa 150g'\n",
      " 'infuzions thai sweetchili potatomix 110g'\n",
      " 'infuzions bbq rib   prawn crackers 110g'\n",
      " 'grnwves plus btroot & chilli jam 180g'\n",
      " 'kettle sweet chilli and sour cream 175g'\n",
      " 'doritos salsa       medium 300g'\n",
      " 'old el paso salsa   dip chnky tom ht300g'\n",
      " 'woolworths mild     salsa 300g'\n",
      " 'old el paso salsa   dip tomato med 300g' 'cheetos puffs 165g'\n",
      " 'rrd sweet chilli &  sour cream 165g' 'woolworths medium   salsa 300g'\n",
      " 'woolworths cheese   rings 190g' 'cheetos chs & bacon balls 190g'\n",
      " 'infuzions sourcream&herbs veg strws 110g'\n",
      " 'infuzions mango     chutny papadums 70g'\n",
      " 'red rock deli chikn&garlic aioli 150g' 'doritos salsa mild  300g']\n",
      "PROD_NAME: 30\n"
     ]
    }
   ],
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```

```
      "39     smiths crinkle cut  french oniondip 150g\n",
      "44        smiths crinkle cut  tomato salsa 150g\n",
      "63              doritos salsa       medium 300g\n",
      "71     old el paso salsa   dip chnky tom ht300g\n",
      "74               woolworths mild     salsa 300g\n",
      "83     smiths crinkle cut  french oniondip 150g\n",
      "87      old el paso salsa   dip tomato med 300g\n",
      "114     old el paso salsa   dip tomato med 300g\n",
      "131     old el paso salsa   dip tomato med 300g\n",
      "149    smiths crinkle cut  french oniondip 150g\n",
      "158               woolworths medium   salsa 300g\n",
      "178    old el paso salsa   dip chnky tom ht300g\n",
      "189    red rock deli sr    salsa & mzzrlla 150g\n",
      "190       smiths crinkle cut  tomato salsa 150g\n",
      "207    old el paso salsa   dip tomato mild 300g\n",
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    "]\n",
    "\n",
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       "       \"grain  waves\",  \"ncc\",  \"red  rock  deli\",  \"dorito\",  \"infuzions\",
\"rrd\", \"infzns\",\n",
    "    \"grnwves\", \"cheetos\"\n",
    "]\n",
    "\n",
    "#Non-chip categories\n",
    "non_chip_keywords = [\n",
    "    \"salsa\",\"dip\",\"nuts\",\"papadums\",\"chocolate\",\"cheese sticks\"\n",
    "]\n",
    "\n",
    "#Lowercase product names\n",
    "prod_lower = df_transactions[\"PROD_NAME\"].str.lower()\n",
    "\n",
    "#Identify likely chips\n",
    "is_chip = prod_lower.str.contains(\"|\".join(chip_keywords), na=False) | \\\n",
     "             prod_lower.str.contains(\"|\".join([b.lower() for b in chip_brands]),
na=False)\n",
    "\n",
```

```
  "# Step 2: Identify obvious non-chips\n",
  "is_non_chip = prod_lower.str.contains(\"|\".join(non_chip_keywords), na=False)\n",
  "\n",
  "# Step 3: Final flag\n",
  "df_transactions[\"is_chip_final\"] = is_chip & (~is_non_chip)\n",
  "\n",
  "# Rows that are definitely not chips\n",
  "df_no_chips = df_transactions[~df_transactions[\"is_chip_final\"]]\n",
  "\n",
  "print(df_no_chips[\"PROD_NAME\"].head(20))\n",
  "print(f\"Total non-chip rows: {len(df_no_chips)}\")"
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    " 1   STORE_NBR       243797 non-null  int64         \n",
    " 2   LYLTY_CARD_NBR  243797 non-null  int64         \n",
    " 3   TXN_ID          243797 non-null  int64         \n",
    " 4   PROD_NBR        243797 non-null  int64         \n",
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    " 7   TOT_SALES       243797 non-null  float64       \n",
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    "75%              2019-03-31 00:00:00     203.000000    2.030860e+05   \n",
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    "\n",
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    "75%    2.026630e+05      86.000000       2.000000       8.800000  \n",
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                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         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    "plt.xlabel('Pack Size (g)')\n",
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       "13                RETIREES       Mainstream    144079.95      37239\n",
       "15          YOUNG FAMILIES           Budget    128575.15      34024\n",
       "9    OLDER SINGLES/COUPLES           Budget    126742.20      32446\n",
       "10   OLDER SINGLES/COUPLES       Mainstream    123643.10      32206\n",
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```

```
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      "3             NEW FAMILIES           Budget     20474.65       5188\n",
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    "14                RETIREES          Premium      23023           3802   \n",
    "2   MIDAGE SINGLES/COUPLES          Premium      14210           2357   \n",
    "0   MIDAGE SINGLES/COUPLES           Budget       8757           1467   \n",
    "13                RETIREES       Mainstream      37239           6326   \n",
    "4             NEW FAMILIES       Mainstream       4028            827   \n",
    "5             NEW FAMILIES          Premium       2745            574   \n",
    "3             NEW FAMILIES           Budget       5188           1086   \n",
    "19   YOUNG SINGLES/COUPLES       Mainstream      35985           7904   \n",
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    "        NUM_CUSTOMERS = ('LYLTY_CARD_NBR','nunique')        \n",
    "    )\n",
    "    .reset_index()\n",
    "    .sort_values(by='TOTAL_QTY', ascending=False)\n",
    ")\n",
    "\n",
              "avg_units[\"AVG_UNITS_PER_CUSTOMER\"]    =    avg_units[\"TOTAL_QTY\"]    /
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      "4             NEW FAMILIES       Mainstream     15899.50     4028   \n",
      "3             NEW FAMILIES           Budget     20474.65     5188   \n",
      "12                RETIREES           Budget    105302.90    26687   \n",
      "14                RETIREES          Premium     90691.65    23023   \n",
      "11   OLDER SINGLES/COUPLES          Premium    122749.15    31379   \n",
      "9    OLDER SINGLES/COUPLES           Budget    126742.20    32446   \n",
      "5             NEW FAMILIES          Premium     10701.00     2745   \n",
      "13                RETIREES       Mainstream    144079.95    37239   \n",
      "10   OLDER SINGLES/COUPLES       Mainstream    123643.10    32206   \n",
      "2   MIDAGE SINGLES/COUPLES          Premium     53973.25    14210   \n",
      "15          YOUNG FAMILIES           Budget    128575.15    34024   \n",
```

```
      "17          YOUNG FAMILIES          Premium     77852.50    20614   \n",
      "0   MIDAGE SINGLES/COUPLES           Budget     33028.30     8757   \n",
      "6           OLDER FAMILIES           Budget    155434.35    41282   \n",
      "7           OLDER FAMILIES       Mainstream     95635.15    25491   \n",
      "16          YOUNG FAMILIES       Mainstream     85547.65    22878   \n",
      "8           OLDER FAMILIES          Premium     74522.60    19950   \n",
      "20   YOUNG SINGLES/COUPLES          Premium     38661.90    10418   \n",
      "18   YOUNG SINGLES/COUPLES           Budget     56514.70    15257   \n",
      "\n",
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    "    .agg(\n",
    "        TOTAL_SALES = ('TOT_SALES','sum'),\n",
    "        PRO_QTY = ('PROD_QTY','sum')\n",
    "    )\n",
    "    .reset_index()\n",
    ")\n",
    "\n",
    "avg_price_per_units[\"AVG_PRICE_PER_UNITS\"] = avg_price_per_units[\"TOTAL_SALES\"] / avg_price_per_units[\"PRO_QTY\"]\n",
    "avg_price_per_units = avg_price_per_units.sort_values(by = \"AVG_PRICE_PER_UNITS\", ascending = False)\n",
    "print(avg_price_per_units)"
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"- \"Young singles/couples\" and \"Midage singles/couples\" pay more per packet of chips compared to their budget counterparts. This might be due to premium shoppers being more likely to buy healthy snacks and when they buy chips, they do it for entertainme/nt purposes rather than their own consumption. "
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"#As the difference in average price per unit isn't large, I would be good to check \n",
"#if the difference is statistically different\n",
"\n",
"#T-test between mainstream vs premium and budget midage, young singles and couples\n",
"\n",
"# Function to perform t-test for a specific lifestage and two premium types\n",
"def ttest_avg_price(df, lifestage, group1, group2):\n",
"   # Filter data for the lifestage and customer groups\n",
"   data1 = df[(df['LIFESTAGE'] == lifestage) & (df['PREMIUM_CUSTOMER'] == group1)]\n",
"   data2 = df[(df['LIFESTAGE'] == lifestage) & (df['PREMIUM_CUSTOMER'] == group2)]\n",

```
    "    \n",
    "    # Calculate price per unit for each transaction\n",
    "    price_per_unit1 = data1['TOT_SALES'] / data1['PROD_QTY']\n",
    "    price_per_unit2 = data2['TOT_SALES'] / data2['PROD_QTY']\n",
    "    \n",
    "    # Perform Welch's t-test (does not assume equal variance)\n",
          "          t_stat,   p_value   =   ttest_ind(price_per_unit1,   price_per_unit2,
equal_var=False)\n",
    "    \n",
    "    print(f\"Lifestage: {lifestage}, Groups: {group1} vs {group2}\")\n",
    "    print(f\"t-statistic = {t_stat:.3f}, p-value = {p_value:.4f}\\n\")\n",
    "\n",
    "# --- MIDAGE SINGLES/COUPLES ---\n",
    "ttest_avg_price(df_merged, 'MIDAGE SINGLES/COUPLES', 'Mainstream', 'Premium')\n",
    "ttest_avg_price(df_merged, 'MIDAGE SINGLES/COUPLES', 'Mainstream', 'Budget')\n",
    "\n",
    "# --- YOUNG SINGLES/COUPLES ---\n",
    "ttest_avg_price(df_merged, 'YOUNG SINGLES/COUPLES', 'Mainstream', 'Premium')\n",
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    "\n",
    "# Filter for Mainstream - YOUNG SINGLES/COUPLES\n",
    "segment_df = df_merged[\n",
    "    (df_merged['LIFESTAGE'] == 'YOUNG SINGLES/COUPLES') &\n",
    "    (df_merged['PREMIUM_CUSTOMER'] == 'Mainstream')\n",
    "]\n",
    "\n",
    "other = df_merged[\n",
    "    ~(\n",
    "        (df_merged['LIFESTAGE'] == 'YOUNG SINGLES/COUPLES') &\n",
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  "quantity_segment1 = segment_df['PROD_QTY'].sum()\n",
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  "\n",
  "# Target segment\n",
  "quantity_segment1_by_brand = (\n",
  "    segment_df.groupby('BRAND')['PROD_QTY']\n",
  "    .sum()\n",
  "    .reset_index()\n",
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  "\n",
  "quantity_segment1_by_brand['targetSegment'] = (\n",
  "    quantity_segment1_by_brand['PROD_QTY'] / quantity_segment1\n",
  ")\n",
  "\n",
  "# Other customers\n",
  "quantity_other_by_brand = (\n",
  "    other.groupby('BRAND')['PROD_QTY']\n",
  "    .sum()\n",
  "    .reset_index()\n",
  ")\n",
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  "quantity_other_by_brand['other'] = (\n",
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  "    on='BRAND'\n",
  ")\n",
  "\n",
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    "6       French Fries       0.003974  0.005830         0.681677\n",
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"# Conclusions\n",
"\n",
"- Sales within the chips category are primarily driven by three key customer segments: Budget ? Older Families, Mainstream ? Young Singles/Couples, and Mainstream ? Retirees. The higher total spend observed among Mainstream Young Singles/Couples and Retirees is largely attributable to their greater representation within the customer base rather than significantly higher individual spending. However, Mainstream customers, particularly mid-age and young singles/couples, also demonstrate a higher willingness to pay per packet, suggesting a tendency towards impulse-driven purchasing behaviour.\n",
"\n",
"- Further analysis reveals that Mainstream Young Singles/Couples show a strong preference for certain brands, being approximately 22% more likely to purchase Tyrrells chips compared to the rest of the population. This indicates a clear opportunity to tailor product placement and promotional strategies towards this segment. \n"
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